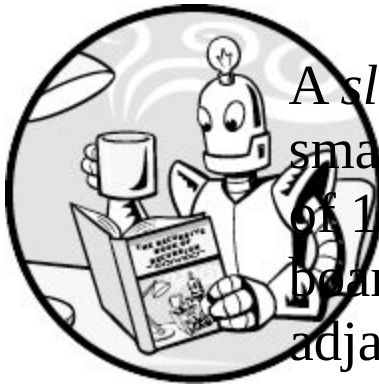
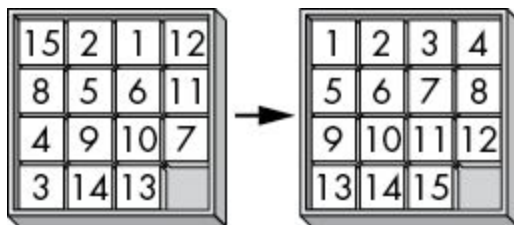


12

SLIDING-TILE SOLVER



A *sliding-tile puzzle*, or *15-puzzle*, is a small puzzle game implemented as a set of 15 numbered sliding tiles on a 4×4 board. One tile is missing, allowing adjacent tiles to slide into the empty space on the board. The player's goal is to move the tiles into numeric order, as in [Figure 12-1](#). Some versions of this game have fragments of a picture on the tiles that create a whole image when the puzzle is complete.



[Figure 12-1](#): Solving a numeric sliding-tile puzzle from its scrambled state (left) to its solved, ordered state (right)

Incidentally, mathematicians have proven that even the hardest 15-puzzle can be solved in 80 moves.